

Compiler/language-infrastructure engineer with professional LLVM and static-analysis experience. In UniversalToolchain I built dependency/conflict resolution, component/provider selection, deterministic pass ordering, typed artifact routing, and pre-execution plan checks; an architecture talk based on the project was accepted for LangDev'26.

2022: Public compiler project · 2023: Language tooling · 2025: Client/product + toolchain · 2026 →: Professional compilers / analysis

EXPERIENCE

- 2026 — present

ISP RAS — Static Analysis Engineer

 - Contribute to SharpChecker, ISP RAS's C#/.NET static-analysis platform.
- July — August 2026

MCST — Compiler Engineering Intern

 - Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
 - Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.
- 2025 — confirmed

Client development — Diploma.school

 - Worked on a real product: jQuery dropdown interactions, responsive mobile/desktop navigation, and UI regression fixes. The work is confirmed no later than January 2025.

PROJECTS

- UniversalToolchain

Language features resolve dependencies and conflicts, select capability/runtime providers, obtain deterministic contribution ordering, and build typed artifact routes before runtime materialization.

Current manifest: 1,324/1,324 tests; dedicated checks cover interpreter/CIL parity and configuration integrity before execution.
- LLVM Interprocedural Global IV Optimization

An LLVM 22 pass localizes supported induction-like globals after interprocedural effect analysis, affine evolution, and a separate applicability check; APInt models fixed-width wraparound.

29 positive/negative regression cases cover LLVM Verifier checks, structural expectations, idempotence, and observable before/after behavior.
- x86-64 Compiler Backend & Register Allocation

SSA/CFG validation → liveness/interference → register allocation/spills → phi lowering → SysV x86-64 emission.

An independent assignment verifier plus native-vs-interpreter differential execution; tests cover branches, loops, spills, and parallel phi moves.

SKILLS

- Compiler Infrastructure

dependency/conflict resolution, component/provider selection, deterministic pass ordering, typed artifact routing, pre-execution checks
- Compiler / IR

C++23, LLVM 22/IR, CFG/SSA, loop and interprocedural analysis, transformation applicability
- Backends / Runtime

.NET, interpreter/CIL parity, SysV x86-64, register allocation, runtime composition
- Verification

regression suites, LLVM Verifier, interpreter/backend parity, differential execution

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

- LangDev'26 — Build the Language, Then Make the Abstractions Disappear

Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga.
- UniversalToolchain — Baltic Science and Engineering Competition

First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain.
- MEPhI Junior

Overall winner; First Degree Diploma, 96/100 for UniversalToolchain.
- Moscow / remote